

Mission 2 Redesign Plan for Replit

Investigation-First Learning Architecture with Selective Mission 1 Immersion Alignment

Executive Summary

Mission 2 should be redesigned from a **command-first networking task** into an **investigation-first cybersecurity learning experience** for beginner users.

The current structure appears to emphasize command execution, output recognition, and answer selection. That approach creates unnecessary cognitive load for beginners because it assumes they already understand what networks, devices, addresses, and inventory relationships mean before they begin investigating.

The redesigned mission should instead lead learners through a structured sequence:

Visual Discovery -> Concept Understanding -> Command Usage -> Evidence Interpretation -> Analyst Judgment

In this model, commands are not the lesson. Commands are **investigation tools** used to answer visible, meaningful questions inside a believable security scenario.

Mission 2 should also selectively adopt the strongest immersive design improvements recently incorporated into Mission 1, but only where those changes strengthen Mission 2's clarity, engagement, and analyst role-play. Mission 2 should not copy Mission 1 wholesale or import design choices that disrupt Mission 2's specific purpose as a beginner-friendly network investigation mission.

The final result should feel less like a Linux command exercise and more like a guided cybersecurity case where the learner discovers an unmanaged contractor device on the finance network, evaluates the risk, and justifies an appropriate response.

Strategic Design Principle

The core principle for Mission 2 should be:

The student should understand why they are using a command before they type it.

The learner should never see `ip addr` and think:

What is this?

They should instead think:

I need to identify my device on the network.

Devices have addresses.

This command should help me find that address.

That shift is the foundation of the redesign.

Mission 2 Design Intent

Mission 2 should teach learners how to:

- understand what devices are on a network
- identify their own workstation
- compare observed assets to approved inventory
- recognize when a device does not belong
- understand why segmentation and ownership matter
- correlate evidence across multiple sources
- make a defensible analyst decision

The mission should not primarily teach:

- memorization of terminal commands
- output pattern matching without understanding
- networking vocabulary without context

The mission should teach **cybersecurity reasoning through guided investigation**.

Mission 1 Alignment Rule

Mission 2 should selectively inherit the immersive design strengths recently introduced in Mission 1, but only as a **supporting presentation layer**.

Mission 2 must preserve its own instructional strategy and gameplay identity.

What Mission 2 may adopt from Mission 1

- stronger environmental storytelling
- better in-world framing
- more cohesive narrative flow
- improved UI transitions

- stronger sense of analyst identity
- more immersive feedback and consequence presentation
- cleaner role-based dialogue and tension

What Mission 2 should not import from Mission 1

- pacing changes that delay or obscure learning
- cinematic flow that replaces evidence-based reasoning
- interaction structures that reduce investigative clarity
- mechanics that overcomplicate beginner concepts
- dramatic layers that interfere with command-to-evidence understanding
- changes that make the mission feel scripted instead of investigative

Mission 1 adaptation principle

Borrow the immersion layer, keep Mission 2's investigation-first strategy.

Core Learning Model

Mission 2 should be structured around this learning loop:

1. **See the environment**
2. **Understand the concept**
3. **Use a command for a reason**
4. **Interpret the evidence**
5. **Update the case**
6. **Make a judgment**
7. **Proceed to the next clue**

This transforms the mission from a terminal exercise into a beginner-friendly analyst simulation.

Redesigned Learning Outcomes

By the end of Mission 2, the learner should be able to:

- explain what a network device is

- explain what an IP address is in practical terms
- identify their workstation on a subnet
- understand why inventories exist
- compare observed assets to approved assets
- recognize why unmanaged devices are risky
- understand why device placement on the wrong segment matters
- connect ownership, location, and behavior into a security conclusion
- recommend a justified response based on evidence

A successful learner should finish Mission 2 thinking:

I discovered that a contractor connected an unmanaged personal device to the finance network and used evidence to justify disconnecting or isolating it.

They should not finish thinking:

I learned ip addr.

Narrative Framing

Mission 2 should open with a clear incident that gives the learner purpose before any command is introduced.

Recommended incident brief

Sarah Reyes, SOC Lead:

We received a monitoring alert from the APAC office. The network inventory says four devices should be present, but the monitoring system sees five. One device may not belong. Before we act, we need to understand the environment and confirm what we're looking at.

This gives the player:

- a clear problem
- a realistic operational reason to investigate
- immediate stakes
- a natural reason to learn the environment first

Mission Flow Overview

Mission 2 should follow a structured multi-stage investigation.

High-level flow

Stage Purpose		Player Activity	Learning Focus
1	Incident framing	Read brief	Why investigation matters
2	Environment introduction	Review network map	Devices exist on a network
3	Concept introduction	Read IP address card	What an address is
4	Self-identification	Run ip addr	Finding a workstation
5	First interpretation	Answer judgment prompt	Meaning of evidence
6	Inventory baseline	Run cat inventory.txt	Approved vs observed
7	Visual comparison	Review board/map	Detect discrepancy
8	Context investigation	Read diagram/assignments	Ownership and segmentation
9	Behavior evidence	Read event log	Suspicious activity
10	Case conclusion	Make determination	Analyst judgment
11	Debrief	Review outcome	Why the decision matters

Detailed Mission Structure

Stage 1: Incident Brief

Goal

Give the player a concrete reason to investigate before any technical interaction begins.

Required experience

The learner enters a live mission context, not a blank command puzzle.

UI behavior

- Mission Brief panel presents the alert

- Sarah Reyes frames the incident
- Objectives appear as investigation tasks, not command tasks

Recommended objectives copy

- Understand the office network environment
- Identify your workstation on the network
- Compare observed devices against approved inventory
- determine whether an unauthorized device is present
- explain why it is risky
- recommend a response

Stage 2: Interactive Network Introduction

Goal

Build a mental model of the network before asking the learner to use commands.

Required experience

The learner should immediately see that:

- multiple devices exist
- devices have addresses
- one device is unresolved
- their own workstation is not yet identified

Recommended visual model

Device	Status	Address
File Server	Approved	192.168.1.10
Finance Laptop	Approved	192.168.1.20
Your Workstation	Unknown	?
Unknown Device	Unverified	???

Replit implementation note

This should live in a persistent network map panel and update over time.

Stage 3: First Learning Pop-Up

Goal

Introduce the idea of an IP address exactly when it becomes useful.

Pop-up content

New Concept: IP Address

A unique network address assigned to a device.

Security teams use IP addresses to identify devices, track activity, and understand what is happening on a network.

Examples:

- File Server -> 192.168.1.10
- Finance Laptop -> 192.168.1.20

Design principle

Keep explanations short, plain-language, and immediately relevant.

Stage 4: First Command Introduction

Goal

Use the first command to answer a meaningful investigative question.

Prompt

Sarah says:

Let's identify your workstation.

Objective

Find your device address.

Player action

Run:

ip addr

Expected output effect

- terminal shows workstation address, such as 192.168.1.34

- map updates Your Workstation: ? -> 192.168.1.34
- notebook logs a new confirmed finding
- confidence indicator increases slightly

Learning outcome

The learner sees that the command answered a question they already understood.

Stage 5: Interpretation Checkpoint

Goal

Require the learner to interpret the evidence, not just read it.

Sample prompt

Analyst Judgment: What did we learn?

- The workstation has a network address
- The workstation is compromised
- The workstation is a server
- Not enough information

Correct reasoning

The evidence confirms the workstation has a network address. It does not prove compromise or server function.

Instructional purpose

This prevents the mission from becoming passive output consumption.

Stage 6: Inventory Discovery

Goal

Show the learner how approved asset baselines are used in security work.

Player action

cat inventory.txt

Required result

The mission reveals the approved list of devices expected in the APAC office.

UI effects

- approved devices panel populates
- notebook logs "Inventory baseline loaded"
- comparison mode unlocks

Learning focus

Inventories exist so analysts can compare what should be present against what is actually present.

Stage 7: Visual Comparison Board

Goal

Make the mismatch immediately visible.

Recommended board state

Approved Devices Observed Devices

File Server	File Server
Finance Laptop	Finance Laptop
Workstation	Workstation
	Unknown Device

Learning outcome

The learner should visually understand the discrepancy before they are asked to explain it.

Design note

This should be one of the strongest visual moments in the mission.

Stage 8: Evidence Correlation

Goal

Shift from simple identification into deeper investigation.

The learner now needs to understand:

- who the device may belong to
- where it is connected

- whether its behavior is normal
- whether it violates policy

Recommended evidence sources

- cat contractor_assignments.txt
- cat network_diagram.txt
- cat security_baseline.txt
- tail network_events.log

Each file should serve a clear investigative role.

Stage 9: Network Diagram Context

Goal

Teach segmentation in a practical way.

Player action

cat network_diagram.txt

Result

The learner sees that:

- finance systems are on an internal segment
- contractor or guest devices should not be there
- placement matters as much as identity

Visual update

The network map should reveal segment boundaries such as:

- Finance Network
- Guest Network
- Corporate LAN

Concept card

Network Segment

A separated portion of a network used to control access and reduce risk.

Learning outcome

The learner understands why a device being on the finance segment matters.

Stage 10: Contractor Ownership Evidence

Goal

Tie the extra device to a plausible human source without immediately normalizing it.

Player action

cat contractor_assignments.txt

Result

The learner finds that a contractor is present in the environment and may be associated with the device.

Important instructional point

Ownership does not equal authorization.

A contractor-owned device may still be:

- unmanaged
- on the wrong network
- violating policy
- high-risk

Judgment prompt

What concerns you most?

- The device belongs to a contractor
- The device is on the finance network
- The device exists on a network
- No concern yet

The strongest answer is the finance network placement, especially if guest access policy exists.

Stage 11: Policy and Baseline Evidence

Goal

Establish what should happen under company rules.

Player action

cat security_baseline.txt

Result

The learner discovers rules such as:

- unmanaged personal devices are not allowed on sensitive internal segments
- contractors must use guest access
- finance network access is restricted

Learning outcome

The learner sees that security decisions are based on policy and environment, not intuition alone.

Stage 12: Behavioral Evidence

Goal

Show why the device is not just unusual but risky.

Player action

tail network_events.log

Result

The unknown device is shown making access attempts toward internal finance resources.

Visual update

Map animates:

- unknown device
- connection path toward finance asset
- warning indicator on sensitive systems

Notebook update

- unapproved device connected internally
- contractor-linked unmanaged device

- wrong segment
- finance access attempts observed

Learning outcome

The learner now has both context and behavioral evidence.

Stage 13: Final Determination

Goal

Require the learner to synthesize all evidence into a professional conclusion.

Determination structure

The learner should answer:

1. Which device is unauthorized?
2. Who likely connected it?
3. Why is it a problem?
4. What evidence supports that conclusion?
5. What response should be taken?

Expected conclusion

- Device: 192.168.1.57
- Likely source: contractor-owned personal laptop
- Problem: unmanaged device connected to finance network
- Supporting evidence: inventory mismatch, contractor records, network segmentation, policy baseline, access attempts
- Recommended response: isolate or disconnect, escalate, document

Success condition

Completion should depend on evidence-backed reasoning, not just selecting the IP.

Stage 14: Debrief

Goal

Lock in the learning after the final determination.

Sarah Reyes debrief should reinforce

- why asset inventories matter
- why segmentation matters
- why unmanaged devices are dangerous
- why ownership does not equal approval
- why evidence supports decisions
- why small policy violations can become serious incidents

End-state feeling

The learner should feel they solved a case.

New Role of Commands

Commands should be explicitly treated as:

Investigation Tools

They should not be framed as:

- trivia to memorize
- isolated objectives
- detached technical hurdles

Command philosophy

Current model:

- Learn ip addr

Redesigned model:

- Need to identify my device
- Use ip addr

That is how real analysts operate.

Required UI and UX Components

1. Live Network Map

This is the central teaching surface.

It should:

- appear before command use begins
- update after meaningful evidence discovery
- reveal devices, segments, relationships, and suspicious behavior
- provide visible progress through investigation

Minimum required updates

- workstation identified
- approved inventory loaded
- extra device highlighted
- finance segment revealed
- contractor relationship shown
- access attempts visualized

2. Analyst Notebook

The notebook should become a structured reasoning tool, not just a passive log.

Recommended notebook sections

Section	Purpose
Confirmed Facts	Evidence directly established
Unknowns	Questions still unresolved
Risks	Why the findings matter
Policy Notes	Relevant organizational rules
Determination	Final case conclusion
Recommended Response Analyst action	

Purpose

Teach learners how investigators organize evidence into a defensible conclusion.

3. Judgment Checkpoints

Every major clue should trigger a short interpretation moment.

These should test

- evidence comprehension
- restraint in making claims
- risk prioritization
- professional reasoning

They should not test

- memorized theory
- obscure networking terms
- syntax mastery without context

4. Just-in-Time Concept Cards

Use small instructional overlays for:

- IP address
- subnet
- inventory
- network segment
- unmanaged device
- baseline policy

These should appear only when needed.

5. Progressive Objective System

Objectives should evolve as evidence is discovered.

Example progression

- Understand the environment
- Identify your workstation
- Load approved inventory
- compare approved and observed devices

- investigate ownership
- assess segment placement
- review activity evidence
- determine response

This keeps the player oriented without overdirecting them.

Replit Technical Implementation Plan

State Model

Mission 2 should be backed by a unified mission state model that drives all panels.

```
type MissionState = {  
  stage: number;  
  identifiedWorkstationIp?: string;  
  approvedInventoryLoaded: boolean;  
  observedDevicesLoaded: boolean;  
  networkDiagramLoaded: boolean;  
  contractorAssignmentsLoaded: boolean;  
  securityBaselineLoaded: boolean;  
  networkEventsLoaded: boolean;  
  discoveredDevices: {  
    id: string;  
    label: string;  
    ip?: string;  
    approved?: boolean;  
    observed?: boolean;  
    segment?: string;  
    owner?: string;  
    managed?: boolean;  
  }  
}
```

```
suspicious?: boolean;
>[];
notebook: {
    confirmedFacts: string[];
    unknowns: string[];
    risks: string[];
    policyNotes: string[];
    determination?: string;
    recommendedResponse?: string;
};
judgmentChecks: {
    id: string;
    completed: boolean;
    correct?: boolean;
>[];
ui: {
    mapState: string;
    confidenceScore: number;
    riskScore: number;
    missionReadiness: number;
};
};
```

Event System

Every meaningful command should trigger structured mission events.

Example event types

```
type MissionEvent =
```

```
| { type: "WORKSTATION_IDENTIFIED"; ip: string }  
| { type: "INVENTORY_LOADED" }  
| { type: "NETWORK_DIAGRAM_LOADED" }  
| { type: "CONTRACTOR_ASSIGNMENTS_LOADED" }  
| { type: "SECURITY_BASELINE_LOADED" }  
| { type: "NETWORK_EVENTS_LOADED" }  
| { type: "JUDGMENT_COMPLETED"; id: string; correct: boolean }  
| { type: "NOTEBOOK_UPDATED" }  
| { type: "FINAL_DETERMINATION_SUBMITTED" };
```

This allows synchronized updates across:

- terminal
- network map
- notebook
- objectives panel
- feedback panel

Command Design Requirements

Commands should be accepted as realistic inputs, but progression should not depend on perfect memorization.

Recommended support

- autocomplete or hinting after failed attempts
- context-sensitive prompting
- command explanation on demand
- recovery from minor syntax mistakes where reasonable

Mission progression rule

Progress should depend on:

- discovering evidence

- interpreting evidence
- forming conclusions

Not just on entering the exact expected string.

Assessment Design

Mission 2 should assess:

- understanding of device identification
- ability to compare approved vs observed assets
- recognition of segment risk
- evidence-based reasoning
- response recommendation quality

Mission 2 should not primarily assess:

- rote command recall
- trivia about networking terminology
- fast guessing

Difficulty and Beginner Accessibility

Mission 2 is for beginners, so the redesign should reduce cognitive overload.

Required beginner protections

- one new concept at a time
- visible cause-and-effect
- short explanations
- map-based reinforcement
- guided interpretation prompts
- recovery from incorrect reasoning
- notebook assistance that scaffolds the case

Important rule

Do not assume prior networking knowledge.

Consequence and Feedback Design

Wrong answers should be corrected through coaching, not harsh failure.

Example feedback model

If the learner says ip addr proves compromise:

Not quite. This confirms the workstation's address on the network, but it does not tell us whether the device is compromised. We need more evidence before making that claim.

This supports learning while preserving seriousness.

Mission 1 Immersive Design Integration Guidance

Mission 2 should selectively incorporate immersive techniques from Mission 1 in the following ways:

Good candidates for reuse

- stronger in-world UI framing
- smoother panel transitions
- improved mission briefing presentation
- better role-based dialogue delivery
- richer feedback when findings are discovered
- stronger consequence framing in debrief moments
- more cohesive analyst identity and operational atmosphere

Elements to avoid copying directly

- interaction pacing that slows investigation
- heavily cinematic sequences that interrupt active reasoning
- systems that obscure evidence flow
- complexity added for dramatic effect
- design changes that turn the mission into narrative consumption instead of analysis

Final immersion rule

Immersion should make the mission feel more real, not less clear.

Success Criteria

Mission 2 is successful if a beginner can finish the mission understanding:

- what a network device is
- what an IP address is
- why inventories are used
- why unauthorized devices are dangerous
- why segment placement matters
- how ownership and policy affect risk
- how evidence supports conclusions
- why the contractor device should be removed or isolated

Final learner takeaway

The learner should be able to say:

I discovered that an unmanaged contractor device was connected to the finance network, observed evidence that it attempted access to sensitive resources, and used inventory, policy, and network context to justify escalation and removal.

Implementation Priorities

Priority 1

Build the **live network map** and connect it to mission state changes.

Priority 2

Restructure the mission around **investigation stages**, not command milestones.

Priority 3

Add **judgment checkpoints** after each major evidence reveal.

Priority 4

Convert the notebook into a **reasoning and conclusion tool**.

Priority 5

Layer in **selective Mission 1 immersive improvements** that strengthen atmosphere without disrupting instructional clarity.

Non-Negotiable Design Rules

1. **Commands must serve visible investigative purposes.**
2. **The network map must update after meaningful discoveries.**
3. **Major evidence must trigger interpretation, not just completion.**
4. **Mission 2 must remain beginner-friendly and concept-first.**
5. **Mission 1 immersion may enhance presentation, but must not replace Mission 2's instructional structure.**
6. **The final success state must require evidence-based judgment, not just identifying one IP address.**

Final Product Statement for Replit

Mission 2 should be redesigned as an investigation-first beginner cybersecurity mission in which learners understand the network environment before being asked to use commands. The mission should guide students through visual discovery, just-in-time concept learning, command-supported evidence gathering, map updates, notebook reasoning, and analyst judgment checkpoints. Commands should function as tools for investigation rather than memorization targets. The redesign should also selectively incorporate the immersive presentation strengths recently added to Mission 1, but only where those changes improve engagement and realism without disrupting Mission 2's clarity, pacing, evidence flow, or detective-style learning experience. The intended outcome is a mission where students do not merely identify an extra device, but understand why an unmanaged contractor laptop on the finance network is risky and how to justify an appropriate response.

Copy-enabled version

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The redesigned mission should instead lead learners through a structured sequence:

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In this model, commands are not the lesson. Commands are investigation tools used to answer visible, meaningful questions inside a believable security scenario.

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The final result should feel less like a Linux command exercise and more like a guided cybersecurity case where the learner discovers an unmanaged contractor device on the finance network, evaluates the risk, and justifies an appropriate response.

Strategic Design Principle

The core principle for Mission 2 should be:

****The student should understand why they are using a command before they type it.****

The learner should never see `ip addr` and think:

> What is this?

They should instead think:

> I need to identify my device on the network.

> Devices have addresses.

> This command should help me find that address.

That shift is the foundation of the redesign.

Mission 2 Design Intent

Mission 2 should teach learners how to:

- understand what devices are on a network

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- stronger sense of analyst identity
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Mission 1 adaptation principle

****Borrow the immersion layer, keep Mission 2's investigation-first strategy.****

Core Learning Model

Mission 2 should be structured around this learning loop:

1. ****See the environment****
2. ****Understand the concept****
3. ****Use a command for a reason****
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6. ****Make a judgment****
7. ****Proceed to the next clue****

This transforms the mission from a terminal exercise into a beginner-friendly analyst simulation.

Redesigned Learning Outcomes

By the end of Mission 2, the learner should be able to:

- explain what a network device is
- explain what an IP address is in practical terms
- identify their workstation on a subnet
- understand why inventories exist

- compare observed assets to approved assets
- recognize why unmanaged devices are risky
- understand why device placement on the wrong segment matters
- connect ownership, location, and behavior into a security conclusion
- recommend a justified response based on evidence

A successful learner should finish Mission 2 thinking:

> I discovered that a contractor connected an unmanaged personal device to the finance network and used evidence to justify disconnecting or isolating it.

They should not finish thinking:

> I learned `ip addr` .

Narrative Framing

Mission 2 should open with a clear incident that gives the learner purpose before any command is introduced.

Recommended incident brief

****Sarah Reyes, SOC Lead:****

> We received a monitoring alert from the APAC office. The network inventory says four devices should be present, but the monitoring system sees five. One device may not belong. Before we act, we need to understand the environment and confirm what we're looking at.

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- a natural reason to learn the environment first

Mission Flow Overview

Mission 2 should follow a structured multi-stage investigation.

High-level flow

| Stage | Purpose | Player Activity | Learning Focus |

|---|---|---|---|

| 1 | Incident framing | Read brief | Why investigation matters |

| 2 | Environment introduction | Review network map | Devices exist on a network |

| 3 | Concept introduction | Read IP address card | What an address is |

| 4 | Self-identification | Run `ip addr` | Finding a workstation |

| 5 | First interpretation | Answer judgment prompt | Meaning of evidence |

| 6 | Inventory baseline | Run `cat inventory.txt` | Approved vs observed |

| 7 | Visual comparison | Review board/map | Detect discrepancy |

| 8 | Context investigation | Read diagram/assignments | Ownership and segmentation |

| 9 | Behavior evidence | Read event log | Suspicious activity |

| 10 | Case conclusion | Make determination | Analyst judgment |

| 11 | Debrief | Review outcome | Why the decision matters |

Detailed Mission Structure

Stage 1: Incident Brief

Goal

Give the player a concrete reason to investigate before any technical interaction begins.

Required experience

The learner enters a live mission context, not a blank command puzzle.

UI behavior

- Mission Brief panel presents the alert
- Sarah Reyes frames the incident
- Objectives appear as investigation tasks, not command tasks

Recommended objectives copy

- Understand the office network environment
- Identify your workstation on the network
- Compare observed devices against approved inventory
- determine whether an unauthorized device is present
- explain why it is risky
- recommend a response

Stage 2: Interactive Network Introduction

Goal

Build a mental model of the network before asking the learner to use commands.

Required experience

The learner should immediately see that:

- multiple devices exist
- devices have addresses
- one device is unresolved
- their own workstation is not yet identified

Recommended visual model

Device	Status	Address
--------	--------	---------

---	---	---
-----	-----	-----

File Server	Approved	192.168.1.10
-------------	----------	--------------

Finance Laptop	Approved	192.168.1.20
----------------	----------	--------------

Your Workstation	Unknown	?
------------------	---------	---

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Replit implementation note

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Goal

Introduce the idea of an IP address exactly when it becomes useful.

Pop-up content

New Concept: IP Address

A unique network address assigned to a device.

Security teams use IP addresses to identify devices, track activity, and understand what is happening on a network.

Examples:

- File Server -> 192.168.1.10

- Finance Laptop -> 192.168.1.20

Design principle

Keep explanations short, plain-language, and immediately relevant.

Stage 4: First Command Introduction

Goal

Use the first command to answer a meaningful investigative question.

Prompt

Sarah says:

> Let's identify your workstation.

Objective

Find your device address.

Player action

Run:

```bash

ip addr

### **Expected output effect**

- terminal shows workstation address, such as 192.168.1.34
- map updates Your Workstation: ? -> 192.168.1.34
- notebook logs a new confirmed finding
- confidence indicator increases slightly

### **Learning outcome**

The learner sees that the command answered a question they already understood.

### **Stage 5: Interpretation Checkpoint**

#### **Goal**

Require the learner to interpret the evidence, not just read it.

#### **Sample prompt**

#### **Analyst Judgment: What did we learn?**

- The workstation has a network address
- The workstation is compromised
- The workstation is a server
- Not enough information

#### **Correct reasoning**

The evidence confirms the workstation has a network address. It does not prove compromise or server function.

#### **Instructional purpose**

This prevents the mission from becoming passive output consumption.

### **Stage 6: Inventory Discovery**

#### **Goal**

Show the learner how approved asset baselines are used in security work.

#### **Player action**

cat inventory.txt

### **Required result**

The mission reveals the approved list of devices expected in the APAC office.

### **UI effects**

- approved devices panel populates
- notebook logs "Inventory baseline loaded"
- comparison mode unlocks

### **Learning focus**

Inventories exist so analysts can compare what should be present against what is actually present.

## **Stage 7: Visual Comparison Board**

### **Goal**

Make the mismatch immediately visible.

### **Recommended board state**

#### **Approved Devices   Observed Devices**

|                |                |
|----------------|----------------|
| File Server    | File Server    |
| Finance Laptop | Finance Laptop |
| Workstation    | Workstation    |
|                | Unknown Device |

### **Learning outcome**

The learner should visually understand the discrepancy before they are asked to explain it.

### **Design note**

This should be one of the strongest visual moments in the mission.

## **Stage 8: Evidence Correlation**

### **Goal**

Shift from simple identification into deeper investigation.



The learner now needs to understand:

- who the device may belong to
- where it is connected
- whether its behavior is normal
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### **Recommended evidence sources**

- `cat contractor_assignments.txt`
- `cat network_diagram.txt`
- `cat security_baseline.txt`
- `tail network_events.log`

Each file should serve a clear investigative role.

## **Stage 9: Network Diagram Context**

### **Goal**

Teach segmentation in a practical way.

### **Player action**

`cat network_diagram.txt`

### **Result**

The learner sees that:

- finance systems are on an internal segment
- contractor or guest devices should not be there
- placement matters as much as identity

### **Visual update**

The network map should reveal segment boundaries such as:

- Finance Network
- Guest Network
- Corporate LAN

## **Concept card**

### **Network Segment**

A separated portion of a network used to control access and reduce risk.

### **Learning outcome**

The learner understands why a device being on the finance segment matters.

## **Stage 10: Contractor Ownership Evidence**

### **Goal**

Tie the extra device to a plausible human source without immediately normalizing it.

### **Player action**

cat contractor\_assignments.txt

### **Result**

The learner finds that a contractor is present in the environment and may be associated with the device.

### **Important instructional point**

Ownership does not equal authorization.

A contractor-owned device may still be:

- unmanaged
- on the wrong network
- violating policy
- high-risk

### **Judgment prompt**

#### **What concerns you most?**

- The device belongs to a contractor
- The device is on the finance network
- The device exists on a network
- No concern yet

The strongest answer is the finance network placement, especially if guest access policy exists.

## **Stage 11: Policy and Baseline Evidence**

### **Goal**

Establish what should happen under company rules.

### **Player action**

cat security\_baseline.txt

### **Result**

The learner discovers rules such as:

- unmanaged personal devices are not allowed on sensitive internal segments
- contractors must use guest access
- finance network access is restricted

### **Learning outcome**

The learner sees that security decisions are based on policy and environment, not intuition alone.

## **Stage 12: Behavioral Evidence**

### **Goal**

Show why the device is not just unusual but risky.

### **Player action**

tail network\_events.log

### **Result**

The unknown device is shown making access attempts toward internal finance resources.

### **Visual update**

Map animates:

- unknown device
- connection path toward finance asset
- warning indicator on sensitive systems

**Notebook update**

- unapproved device connected internally
- contractor-linked unmanaged device
- wrong segment
- finance access attempts observed

**Learning outcome**

The learner now has both context and behavioral evidence.

**Stage 13: Final Determination****Goal**

Require the learner to synthesize all evidence into a professional conclusion.

**Determination structure**

The learner should answer:

1. Which device is unauthorized?
2. Who likely connected it?
3. Why is it a problem?
4. What evidence supports that conclusion?
5. What response should be taken?

**Expected conclusion**

- Device: 192.168.1.57
- Likely source: contractor-owned personal laptop
- Problem: unmanaged device connected to finance network
- Supporting evidence: inventory mismatch, contractor records, network segmentation, policy baseline, access attempts
- Recommended response: isolate or disconnect, escalate, document

**Success condition**

Completion should depend on evidence-backed reasoning, not just selecting the IP.

## **Stage 14: Debrief**

### **Goal**

Lock in the learning after the final determination.

### **Sarah Reyes debrief should reinforce**

- why asset inventories matter
- why segmentation matters
- why unmanaged devices are dangerous
- why ownership does not equal approval
- why evidence supports decisions
- why small policy violations can become serious incidents

### **End-state feeling**

The learner should feel they solved a case.

### **New Role of Commands**

Commands should be explicitly treated as:

### **Investigation Tools**

They should not be framed as:

- trivia to memorize
- isolated objectives
- detached technical hurdles

### **Command philosophy**

Current model:

- Learn ip addr

Redesigned model:

- Need to identify my device
- Use ip addr

That is how real analysts operate.

## Required UI and UX Components

### 1. Live Network Map

This is the central teaching surface.

#### It should:

- appear before command use begins
- update after meaningful evidence discovery
- reveal devices, segments, relationships, and suspicious behavior
- provide visible progress through investigation

#### Minimum required updates

- workstation identified
- approved inventory loaded
- extra device highlighted
- finance segment revealed
- contractor relationship shown
- access attempts visualized

### 2. Analyst Notebook

The notebook should become a structured reasoning tool, not just a passive log.

#### Recommended notebook sections

| Section                             | Purpose                       |
|-------------------------------------|-------------------------------|
| Confirmed Facts                     | Evidence directly established |
| Unknowns                            | Questions still unresolved    |
| Risks                               | Why the findings matter       |
| Policy Notes                        | Relevant organizational rules |
| Determination                       | Final case conclusion         |
| Recommended Response Analyst action |                               |

## **Purpose**

Teach learners how investigators organize evidence into a defensible conclusion.

### **3. Judgment Checkpoints**

Every major clue should trigger a short interpretation moment.

#### **These should test**

- evidence comprehension
- restraint in making claims
- risk prioritization
- professional reasoning

#### **They should not test**

- memorized theory
- obscure networking terms
- syntax mastery without context

### **4. Just-in-Time Concept Cards**

Use small instructional overlays for:

- IP address
- subnet
- inventory
- network segment
- unmanaged device
- baseline policy

These should appear only when needed.

### **5. Progressive Objective System**

Objectives should evolve as evidence is discovered.

#### **Example progression**

- Understand the environment

- Identify your workstation
- Load approved inventory
- compare approved and observed devices
- investigate ownership
- assess segment placement
- review activity evidence
- determine response

This keeps the player oriented without overdirecting them.

## **Replit Technical Implementation Plan**

### **State Model**

Mission 2 should be backed by a unified mission state model that drives all panels.

```
type MissionState = {
 stage: number;
 identifiedWorkstationIp?: string;
 approvedInventoryLoaded: boolean;
 observedDevicesLoaded: boolean;
 networkDiagramLoaded: boolean;
 contractorAssignmentsLoaded: boolean;
 securityBaselineLoaded: boolean;
 networkEventsLoaded: boolean;
 discoveredDevices: {
 id: string;
 label: string;
 ip?: string;
 approved?: boolean;
 observed?: boolean;
```



```
 segment?: string;
 owner?: string;
 managed?: boolean;
 suspicious?: boolean;
 }[];
 notebook: {
 confirmedFacts: string[];
 unknowns: string[];
 risks: string[];
 policyNotes: string[];
 determination?: string;
 recommendedResponse?: string;
 };
 judgmentChecks: {
 id: string;
 completed: boolean;
 correct?: boolean;
 }[];
 ui: {
 mapState: string;
 confidenceScore: number;
 riskScore: number;
 missionReadiness: number;
 };
};
```

## **Event System**

Every meaningful command should trigger structured mission events.

### **Example event types**

type MissionEvent =

```
| { type: "WORKSTATION_IDENTIFIED"; ip: string }
| { type: "INVENTORY_LOADED" }
| { type: "NETWORK_DIAGRAM_LOADED" }
| { type: "CONTRACTOR_ASSIGNMENTS_LOADED" }
| { type: "SECURITY_BASELINE_LOADED" }
| { type: "NETWORK_EVENTS_LOADED" }
| { type: "JUDGMENT_COMPLETED"; id: string; correct: boolean }
| { type: "NOTEBOOK_UPDATED" }
| { type: "FINAL_DETERMINATION_SUBMITTED" };
```

This allows synchronized updates across:

- terminal
- network map
- notebook
- objectives panel
- feedback panel

### **Command Design Requirements**

Commands should be accepted as realistic inputs, but progression should not depend on perfect memorization.

### **Recommended support**

- autocomplete or hinting after failed attempts
- context-sensitive prompting
- command explanation on demand
- recovery from minor syntax mistakes where reasonable

### **Mission progression rule**

Progress should depend on:

- discovering evidence
- interpreting evidence
- forming conclusions

Not just on entering the exact expected string.

### **Assessment Design**

Mission 2 should assess:

- understanding of device identification
- ability to compare approved vs observed assets
- recognition of segment risk
- evidence-based reasoning
- response recommendation quality

Mission 2 should not primarily assess:

- rote command recall
- trivia about networking terminology
- fast guessing

### **Difficulty and Beginner Accessibility**

Mission 2 is for beginners, so the redesign should reduce cognitive overload.

### **Required beginner protections**

- one new concept at a time
- visible cause-and-effect
- short explanations
- map-based reinforcement
- guided interpretation prompts
- recovery from incorrect reasoning

- notebook assistance that scaffolds the case

### **Important rule**

Do not assume prior networking knowledge.

### **Consequence and Feedback Design**

Wrong answers should be corrected through coaching, not harsh failure.

### **Example feedback model**

If the learner says ip addr proves compromise:

Not quite. This confirms the workstation's address on the network, but it does not tell us whether the device is compromised. We need more evidence before making that claim.

This supports learning while preserving seriousness.

### **Mission 1 Immersive Design Integration Guidance**

Mission 2 should selectively incorporate immersive techniques from Mission 1 in the following ways:

#### **Good candidates for reuse**

- stronger in-world UI framing
- smoother panel transitions
- improved mission briefing presentation
- better role-based dialogue delivery
- richer feedback when findings are discovered
- stronger consequence framing in debrief moments
- more cohesive analyst identity and operational atmosphere

#### **Elements to avoid copying directly**

- interaction pacing that slows investigation
- heavily cinematic sequences that interrupt active reasoning
- systems that obscure evidence flow
- complexity added for dramatic effect
- design changes that turn the mission into narrative consumption instead of analysis

## **Final immersion rule**

Immersion should make the mission feel more real, not less clear.

## **Success Criteria**

Mission 2 is successful if a beginner can finish the mission understanding:

- what a network device is
- what an IP address is
- why inventories are used
- why unauthorized devices are dangerous
- why segment placement matters
- how ownership and policy affect risk
- how evidence supports conclusions
- why the contractor device should be removed or isolated

## **Final learner takeaway**

The learner should be able to say:

I discovered that an unmanaged contractor device was connected to the finance network, observed evidence that it attempted access to sensitive resources, and used inventory, policy, and network context to justify escalation and removal.

## **Implementation Priorities**

### **Priority 1**

Build the live network map and connect it to mission state changes.

### **Priority 2**

Restructure the mission around investigation stages, not command milestones.

### **Priority 3**

Add judgment checkpoints after each major evidence reveal.

### **Priority 4**

Convert the notebook into a reasoning and conclusion tool.

### **Priority 5**

Layer in selective Mission 1 immersive improvements that strengthen atmosphere without disrupting instructional clarity.

### **Non-Negotiable Design Rules**

1. **Commands must serve visible investigative purposes.**
2. **The network map must update after meaningful discoveries.**
3. **Major evidence must trigger interpretation, not just completion.**
4. **Mission 2 must remain beginner-friendly and concept-first.**
5. **Mission 1 immersion may enhance presentation, but must not replace Mission 2's instructional structure.**
6. **The final success state must require evidence-based judgment, not just identifying one IP address.**

### **Final Product Statement for Replit**

Mission 2 should be redesigned as an investigation-first beginner cybersecurity mission in which learners understand the network environment before being asked to use commands. The mission should guide students through visual discovery, just-in-time concept learning, command-supported evidence gathering, map updates, notebook reasoning, and analyst judgment checkpoints. Commands should function as tools for investigation rather than memorization targets. The redesign should also selectively incorporate the immersive presentation strengths recently added to Mission 1, but only where those changes improve engagement and realism without disrupting Mission 2's clarity, pacing, evidence flow, or detective-style learning experience. The intended outcome is a mission where students do not merely identify an extra device, but understand why an unmanaged contractor laptop on the finance network is risky and how to justify an appropriate response.

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